Flexible-margin Gaussian-copula full random-effects models: likelihood estimation and posterior-based margin selection

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Abstract

Full random-effects modelling (FREM) jointly describes individual parameters and covariates using a multivariate Normal population distribution. We extend FREM by combining Gaussian dependence with flexible continuous marginal distributions for parameter random effects and continuous or ordered categorical margins for covariates. Because parameter random effects are latent, their marginal families cannot be selected directly from the observed data. We therefore propose a screen-and-refit likelihood workflow: a Normal-eta incumbent supplies full individual posterior samples, candidate eta families are ranked on independent training and validation pools, and the selected families are frozen before a fresh fit. For each fixed family specification, population locations, marginal parameters, dependence parameters and residual error are estimated jointly by controlled-Markov likelihood-score stochastic approximation. The model reduces to Gaussian FREM when all margins are Normal and permits missing covariates. Posterior and direct observed-likelihood family rankings agreed in all 24 informative datasets and in 22 of 24 weak-information datasets with 2,000 draws; increasing the two unresolved comparisons to 20,000 draws recovered agreement. In 400 simulated nonlinear mixed-effects datasets, structural-parameter bias remained within 0.47%. Relative to Gaussian FREM, mean differences in observed log likelihood were 4.30–9.41 under non-Gaussian data-generating distributions and -0.06 under a Gaussian data-generating distribution. Among In 200 independent high-CRP prediction datasets, flexible margins had lower VPC log-RMSE in 174; median RMSE was 0.085 versus 0.336 for Gaussian FREM, and the mean paired difference was 0.246 (95% CI 0.219–0.272). These results indicate that latent margin selection can be separated from fixed-model likelihood estimation and that marginal assumptions can materially affect conditional prediction. The fitted conditional model has an exact conventional FFEM representation when parameter margins are Normal and an exact conditional generative representation otherwise.

Keywords: nonlinear mixed-effects model; full random-effects model; Gaussian copula; stochastic approximation; marginal likelihood; covariates

INTRODUCTION

Covariate models describe how patient characteristics contribute to variability in individual model parameters and predictions. Full random-effects modelling (FREM) includes a prespecified set of covariates as additional random effects and estimates their joint covariance with the parameter random effects [1]. Covariate-effect coefficients and unexplained parameter variability can then be calculated for any subset of the included covariates from a single estimation. This avoids repeated estimation of separate reduced covariate models and allows missing covariates to be integrated over in the likelihood [2, 3].

Standard FREM assumes that the combined parameter and covariate random effects are multivariate Normal. For covariates observed with negligible error, this working model can estimate means and covariances accurately under some non-Normal covariate distributions [1]. Recent FREM investigations also show that covariate-model misspecification can affect inclusion and omission bias [4]. The Gaussian working model nevertheless specifies the entire population distribution, including marginal shape and conditional tail behaviour. These features are relevant when the fitted model is used for conditional prediction, simulation of virtual populations, integration over missing covariates, or extrapolation to sparsely represented covariate values.

A copula permits the marginal distributions to be specified separately from their dependence [5, 6]. We extend FREM by combining a Gaussian copula with flexible continuous margins for parameter random effects and continuous or ordered categorical margins for covariates. The multivariate Normal FREM model is recovered when all margins are Normal. Continuous covariate margins can be screened directly from their observed values, whereas the eta margins cannot because the individual random effects are latent. Selecting their families from empirical Bayes estimates would ignore individual uncertainty and shrinkage.

The proposed workflow therefore separates discrete family selection from continuous likelihood estimation. A Normal-eta incumbent is fitted first. Full draws from its individual conditional distributions are then used to screen candidate eta families by an exact observed-likelihood ratio identity. The selected families are frozen, and all continuous population and residual-error parameters are re-estimated in a fresh fit. At no point are eta families changed inside the stochastic-approximation recursion.

The objectives of this work were to define the flexible-margin FREM model and its observed-data likelihood, establish a likelihood-score estimator for any fixed admissible margin specification, develop posterior-based automatic eta-margin selection, derive the corresponding conditional and FFEM representations, and evaluate estimation and conditional prediction in simulation studies.

METHODS

Statistical model

Gaussian FREM

For subject i at design point x_{ij}, write the nonlinear mixed-effects observation model as [7]

$$y_{ij} = f(\psi_i, x_{ij}) + g(\psi_i, \sigma, x_{ij})\epsilon_{ij}, \qquad \epsilon_{ij} \sim N(0,1), \tag{1}$$

where ψ_i is the vector of individual model parameters and σ contains the residual-error parameters. On the transformed additive scale,

$$\phi_i = h(\psi_i) = C_i\mu + \eta_{\text{par},i}, \tag{2}$$

where C_i is the fixed-effect design matrix, μ is the vector of population parameters, and $\eta_{\text{par},i}$ is the parameter random effect. The lowercase symbol c_i denotes the FREM covariate vector and distinguishes it from the design matrix C_i. An error-free additive FREM covariate model is

$$c_i = \mu_c + \eta_{\text{cov},i}. \tag{3}$$

Here $\eta_{\text{par},i}$ represents total parameter variability (TPV) and $\eta_{\text{cov},i}$ is the individual covariate deviation. Standard FREM specifies

$$\begin{pmatrix} \eta_{\text{par},i} \\ \eta_{\text{cov},i} \end{pmatrix} \sim N\left[0, \Omega_{\text{FREM}}\right], \qquad \Omega_{\text{FREM}} = \begin{pmatrix} \Omega_{\text{par}} & \Omega_{\text{par,cov}} \\ \Omega_{\text{cov,par}} & \Omega_{\text{cov}} \end{pmatrix}. \tag{4}$$

Equation (4) is the standard FREM specification [8, 1]. Let $\Omega_{\text{par,cov}} \in \mathbb{R}^{p\times q}$ and $\Omega_{\text{cov,par}} = \Omega_{\text{par,cov}}^{\mathsf{T}}$. The covariate-effect matrix and unexplained parameter variability (UPV) are

$$B = \Omega_{\text{par,cov}}\Omega_{\text{cov}}^{-1}, \qquad \Omega'_{\text{par}} = \Omega_{\text{par}} - B\Omega_{\text{cov,par}}. \tag{5}$$

Consequently,

$$\eta_{\text{par},i} \mid c_i \sim N\left[B(c_i - \mu_c), \Omega'_{\text{par}}\right]. \tag{6}$$

Equations (4)–(6) give the standard Gaussian FREM model, its conditional covariate coefficients, and its UPV representation.

Flexible-margin Gaussian-copula FREM

Let $X_i = (\eta_{\text{par},i}, c_i)$ contain the d population coordinates. For continuous coordinate j, write $F_j(\cdot;\alpha_j)$ and $f_j(\cdot;\alpha_j)$ for its CDF and density and define

$$\xi_j = \Phi^{-1}\{F_j(x_j;\alpha_j)\}.$$

Let $R(\rho)$ be a positive-definite Gaussian-score correlation matrix and let $\theta_p = (\alpha, \rho)$ collect the population-distribution parameters. The continuous population density is

$$p_{\theta_p}(x) = |R|^{-1/2} \exp\big[-\tfrac{1}{2}\xi^\mathsf{T}(R^{-1} - I)\xi\big] \prod_{j=1}^{d} f_j(x_j;\alpha_j). \tag{7}$$

We parameterize R by a fixed regular vine with Gaussian pair copulas; the vine structure is not estimated during the stochastic-approximation recursion. When all margins are Normal on their modelled scales, (7) reduces exactly to the error-free Gaussian FREM population model (2)–(4), with Ω_{FREM} obtained from the marginal standard deviations and R. This error-free formulation is the zero-covariate-residual limit of the conventional NONMEM/PsN formulation, which uses a small fixed residual variance for covariate responses [1, 3].

Let $\theta = (\mu, \alpha, \rho, \sigma)$ denote the unknown parameter vector. Here α contains the parameters of the marginal distributions of η_{par} and c, including μ_c; ρ parameterizes R; and σ contains the residual-error parameters. When all margins are Normal, the marginal scale parameters together with ρ are equivalently represented by $\text{vec}(\Omega_{\text{FREM}})$; the covariate locations μ_c remain separate.

The corresponding FREM effect summaries live naturally on the latent-score scale. Partition R into individual-parameter and covariate blocks. Then

$$B_\xi = R_{\text{par,cov}} R_{\text{cov}}^{-1}, \qquad R'_{\text{par}} = R_{\text{par}} - B_\xi R_{\text{cov,par}}, \tag{7a}$$

so that $\xi_{\text{par}} \mid c$ is Normal with mean $B_\xi \xi_{\text{cov}}(c)$ and covariance R'_{par}. The fitted marginal quantile maps transform this conditional law back to the native parameter scale. Under all-Normal margins, let D_{par} and D_{cov} be diagonal matrices of marginal standard deviations. Then $B = D_{\text{par}} B_\xi D_{\text{cov}}^{-1}$ and $\Omega'_{\text{par}} = D_{\text{par}} R'_{\text{par}} D_{\text{par}}$, which recovers (5). Under non-Gaussian parameter margins the native-scale conditional mean need not be linear. In that case, B_ξ, R'_{par} and the marginal quantile functions jointly determine the conditional distribution.

Estimation and margin-selection workflow

The complete workflow contains four steps. First, continuous covariate margins are either prespecified or screened by marginal AIC from Normal, lognormal, Gamma and Weibull candidates using their observed values; the selected families are then fixed. Categorical margins and their ordering are declared from their scientific meaning. Second, a Normal-eta Gaussian-copula incumbent is fitted to the joint observed-data likelihood. This is ordinary Gaussian FREM only when the covariate margins are also Normal. Third, full draws from the incumbent individual conditional distributions are used to rank candidate eta-margin families. Fourth, the selected family combination is frozen and a fresh fixed-family model is fitted, re-estimating its population, dependence and residual-error parameters. A user may bypass screening by prespecifying scientifically justified margins.

This separation is essential: the fixed-family estimator has one unchanged state space and likelihood throughout its recursion, while family selection is a finite outer model-comparison step. The convergence result below concerns the incumbent and final fixed-family fits, not the discrete selection operation.

Observed-data likelihood

Let $O_i = (Y_i, c_{i,\mathrm{obs}})$ and let Z_i collect individual parameters and missing covariates. For categorical covariates and continuous margins with parameter-dependent support, Z_i additionally contains the fixed-support unit-interval variables defined below. With a fixed mixed Lebesgue/counting measure λ, define the complete-data likelihood f_i and observed-data likelihood g_i by

$$g_i(\theta) = \int f_i(z_i;\theta)\,\lambda(dz_i), \qquad \ell(\theta) = \sum_{i=1}^{N} \log g_i(\theta). \tag{8}$$

The dependence of f_i and g_i on O_i is implicit, as in Delyon et al.; Z_i denotes the missing or augmented data [9]. Consequently, the likelihood contribution is based on the joint density $p_\theta(Y_i, c_{i,\mathrm{obs}})$. Individual effects and missing covariates are integrated out rather than replaced by point estimates. The analysis conditions on the recorded missingness pattern and assumes that the covariate-missingness mechanism is ignorable.

When all covariates are observed, its subject contribution has the exact factorization

$$\log p_\theta(Y_i, c_i) = \log p_{\theta_p}(c_i) + \log p_\theta(Y_i \mid c_i). \tag{8a}$$

Thus a difference in joint log likelihood may reflect differences in the fitted covariate density, the conditional response likelihood, or both. This factorization does not alter the estimation objective in (8).

When the continuous margins are proper and absolutely continuous, the required fixed-support representations exist, the categorical probability vectors are proper, R is positive definite, and the response density is proper, (7) defines a probability distribution and $g_i(\theta)$ in (8) is a valid observed-data likelihood contribution. This follows from a componentwise probability-integral transformation: continuous Jacobians transform (7) to a $N_d(0, R)$ density, categorical cells partition their latent Gaussian coordinates, and the fixed-support representations put moving boundaries on fixed unit intervals. Appendix A gives the detailed normalization argument.

Fixed-family estimation problem

For the Gaussian FREM model, the complete-data likelihood has the curved exponential-family representation used by standard SAEM,

$$\log p_\theta(Y, Z) = -\psi(\theta) + \langle S(Y, Z), \varphi(\theta)\rangle,$$

where $S(Y, Z)$ is a parameter-independent finite-dimensional sufficient statistic. Standard SAEM recursively averages $S(Y, Z)$ and performs the M-step from that finite vector [9, 10].

In (7), a marginal parameter α_j enters both $\log f_j(x_j; \alpha_j)$ and the transformed value $\xi_j = \Phi^{-1}\{F_j(x_j; \alpha_j)\}$ inside the copula quadratic form. Marginal and dependence parameters are therefore coupled through nonlinear functions of the latent data, and a parameter-independent finite statistic $S(Y, Z)$ is not generally available. The observed-data likelihood and the EM identity remain valid, but the usual finite-statistic SAEM recursion and its convergence theorem cannot simply be carried over; the curved-exponential condition is a recognized bottleneck for SAEM [11]. One possible workaround adds fixed-variance Gaussian latent variables to exponentialize the model, but then maximizes a perturbed likelihood whose approximation and numerical behaviour depend on that variance [12, 11]. Here the likelihood score of the original, unperturbed model is used instead, through the stochastic-gradient recursion proposed by Younes and presented as equation (74) by Delyon et al. [13, 9]. When all margins are Normal, the ordinary Gaussian FREM finite-statistic representation is recovered.

Controlled-Markov approximation of the likelihood score

Work in unconstrained internal coordinates ϑ, with native parameters $\theta = \tau(\vartheta)$. Log, logit and hyperspherical transformations enforce positive scales, probabilities and a positive-definite R. Write $\widetilde{f}_i(z;\vartheta) = f_i\{z;\tau(\vartheta)\}$ and $\widetilde{g}_i(\vartheta) = g_i\{\tau(\vartheta)\}$. At iteration k, controlled Markov kernels invariant for the current individual conditional law advance m chains per subject. The estimated complete score is

$$\widehat{H}_{k+1} = \frac{1}{mN}\sum_{r=1}^{m}\sum_{i=1}^{N}\nabla_\vartheta \log f_i(Z_{i,k+1}^{(r)};\tau(\vartheta_k)), \tag{9}$$

and the parameter update is

$$\vartheta_{k+1} = \mathrm{Proj}_K\{\vartheta_k + \gamma_{k+1}A_k\widehat{H}_{k+1}\}. \tag{10}$$

Adaptation of the positive diagonal metric and proposal scales is restricted to a finite initial sequence; these quantities are subsequently held fixed. The gain sequence is $\gamma_k = \gamma_0(k+k_0)^{-a}$ with $3/4 < a \leq 1$. Marginal families, category order, copula structure and all other discrete model choices are specified before recursion (10) begins.

Equation (10) is a controlled-Markov, internal-coordinate, projected, preconditioned and multiple-chain extension of the Younes stochastic-gradient recursion presented as Delyon et al. equation (74), $\theta_k = \theta_{k-1} + \gamma_k\partial_\theta \log f(Z_k;\theta_{k-1})$. The coordinate transformations and fixed positive-definite metric leave the stationary points unchanged. Because the implemented chains persist between iterations, their draws are Markovian and do not satisfy the conditional-independence assumption used in Delyon et al.'s direct analysis of equation (74). Appendix B replaces that martingale-difference argument by the Poisson-equation decomposition for controlled-Markov stochastic approximation, building on the SAEM–MCMC construction and its later extensions [14, 15].

Every stochastic move is either an exact conditional draw or a Metropolis- Hastings transition with the current individual conditional law as invariant distribution. At least one exact conditional-population independence proposal is used per iteration. With several categorical covariates, random-walk moves that would require a numerical rectangle probability are omitted; the exact independence kernel remains. An unadjusted approximate kernel would generally have a different invariant distribution and hence a different mean field [16].

Likelihood target

Provided differentiation can pass through (8), Fisher's identity gives

$$E_\vartheta\{\nabla_\vartheta \log \widetilde{f}_i(Z_i;\vartheta) \mid O_i\} = \nabla_\vartheta \log \widetilde{g}_i(\vartheta). \tag{11}$$

For one subject, substitution of the conditional density $\widetilde{f}_i(z;\vartheta)/\widetilde{g}_i(\vartheta)$ into the expectation reduces its numerator to $\nabla_\vartheta \int \widetilde{f}_i(z;\vartheta)\lambda(dz)$. The fixed-support representations ensure that λ is fixed. Appendix A states the required domination conditions and gives the full calculation [17].

Let $\bar{\ell} = N^{-1}\ell$. Thus the mean field after finite tuning is $A\nabla\bar{\ell}$. For symmetric positive-definite A,

$$\frac{d}{dt}\bar{\ell}\{\vartheta(t)\} = \nabla\bar{\ell}(\vartheta)^\mathsf{T} A\nabla\bar{\ell}(\vartheta) \geq 0. \tag{12}$$

Convergence

Let $\mathcal{L} = \{\vartheta : \nabla\bar{\ell}(\vartheta) = 0\}$. Suppose the fixed model is continuously differentiable on the stabilized compact interior set; differentiation is dominated; the controlled kernels have unique invariant laws, uniform geometric drift/minorization and the local Poisson-equation regularity; the gain satisfies the Fort condition; adaptation is finite and numerical error is gain-weighted summable; safeguards eventually cease; and $-\bar{\ell}$ has the required coercive Lyapunov extension and critical-value property. Then controlled-Markov stochastic approximation gives

$$d(\vartheta_k, \mathcal{L}) \longrightarrow 0 \qquad \text{almost surely.} \tag{13}$$

The limit set lies in a connected component of $\mathcal{L}$ and collapses to an isolated attracting point when one contains the tail. This is a conditional stationary-set result, not a global-maximum claim and not an application of the finite-statistic SAEM Theorems 5 or 8. Appendix B states and maps the complete assumptions [18, 15].

Posterior-based automatic eta-margin selection

Eta-margin families are discrete model choices and are not changed during recursion (10). They may instead be screened before a fresh fixed-model fit by reusing the complete individual posterior distribution from a fitted Normal-eta incumbent. Let f_{0i} and g_{0i} denote the incumbent complete- and observed-data likelihoods, and let $\pi_{0i}(z) = f_{0i}(z)/g_{0i}$ be its individual conditional distribution. For a candidate model m with the same observation model, population locations, covariate margins and Gaussian dependence during screening,

$$\frac{g_{mi}}{g_{0i}} = E_{\pi_{0i}} \left\{ \frac{f_{mi}(Z_i)}{f_{0i}(Z_i)} \right\} = E_{\pi_{0i}} \left\{ \frac{p_m(\eta_i, c_i)}{p_0(\eta_i, c_i)} \right\}. \tag{14}$$

Unlike EBEs, these draws retain individual uncertainty. Candidate parameters are optimized on one posterior pool; an independently generated pool gives the validation BIC advantage

$$2 \sum_{i=1}^{N} \log \left[\frac{1}{M} \sum_{r=1}^{M} \frac{p_m(\eta_i^{(r)}, c_i)}{p_0(\eta_i^{(r)}, c_i)} \right] - (k_m - k_0) \log N, \tag{15}$$

relative to the incumbent. The screened set is centred Normal, Student and Laplace, with incumbent dependence and non-eta margins held fixed. MCMC ESS, importance ESS, batch-means Monte Carlo error and ranking separation determine whether the result is resolved. The log of the importance average has conservative Jensen bias toward the incumbent, and Normal-to-Student weights can have infinite variance; empirical ESS detects concentration but cannot prove a finite second moment. Unresolved screens require more draws, fewer candidates or direct comparison of refitted leaders.

The selected families are frozen for a fresh score fit that re-estimates all continuous population, dependence and residual-error parameters. Thus (13) applies to that fixed-model fit, not to selection itself. Appendix A proves (14) and gives the estimator qualifications.

Conditional representation and FFEM translation

For a selected covariate subset S, let z_{iS} contain the corresponding Gaussian scores and calculate $B_{\xi,S}$ and $R'_{\text{par},S}$ from the relevant blocks of R. If the parameter random- effect margins are Normal, the exact FFEM is

$$\phi_i = C_i\mu + \beta_{z,S} z_{iS} + \eta'_{iS}, \qquad \beta_{z,S} = D_{\text{par}} B_{\xi,S}, \qquad \eta'_{iS} \sim N(0, \Omega'_{\text{par},S}),$$

where $\Omega'_{\text{par},S} = D_{\text{par}} R'_{\text{par},S} D_{\text{par}}$. Covariates are fixed inputs whose fitted CDFs produce z_{iS}; all-Normal covariate margins recover the usual raw-scale FREM formulas [1].

For non-Normal parameter random effects, a conventional additive-Normal FFEM is not generally equivalent. Let Q_η apply the fitted parameter-margin quantile functions componentwise and choose L_S such that $L_S L_S^\top = R'_{\text{par},S}$. The exact conditional representation is

$$\phi_i = C_i\mu + Q_\eta[\Phi\{B_{\xi,S} z_{iS} + L_S \varepsilon_i\}], \qquad \varepsilon_i \sim N(0, I).$$

This transformed latent-Normal law retains the non-Normal margins exactly; it is a conditional generative representation, not a conventional additive-Normal FFEM. Ordered categories use the corresponding latent-interval mixture. The translation represents $p(y \mid c_S)$ and therefore omits the covariate-density term in the joint FREM likelihood. Appendix A proves both cases.

Categorical covariates and parameter-dependent support

Binary, ordinal and explicitly ordered nominal covariates are represented by their latent Gaussian intervals; continuous moving-support coordinates use a fixed percentile and the candidate quantile map. Both augmentations live on a parameter-independent reference space, so Fisher's identity remains applicable and multiple categorical coordinates are allowed. Permutation-invariant nominal models and nonregular support-boundary MLEs remain outside scope. Appendix A gives the constructions and normalization.

Simulation and numerical evaluation

Fixed-family re-estimation

The primary simulation isolated fixed-family estimation from family selection. Four scenarios (Normal/Normal, lognormal/Normal, Gamma/Normal and Weibull/lognormal covariates) each contained 100 independently generated one-compartment datasets with $N = 100$, two eta coordinates and two covariates. Generating families were supplied to the flexible fit. Each fit used 1,500 score iterations with persistent invariant kernels and gain exponent 0.80; all adaptation stopped after 50 iterations. Gaussian and flexible fits were reevaluated with independent 5,000-draw importance samples. The complete parameters, dependence matrix, seeds and stabilization criteria are recorded in the reproducibility bundle.

Automatic eta-margin selection

The posterior identity was evaluated in an additive latent model, $Y_i = \eta_i + \epsilon_i$, where the direct observed likelihood is available analytically for Normal and Laplace eta margins. Twelve datasets per generating family were simulated with $N = 100$, eta standard deviation 0.45 and residual standard deviation 0.25. A fitted Normal-eta incumbent supplied 2,000 posterior draws per subject for each of two independently generated pools. Candidate scale parameters were fitted on the first pool and family rankings were evaluated on the second. A weak-information setting used residual standard deviation 0.60; discordant close comparisons were repeated with 20,000 draws per subject, providing an exact likelihood benchmark independent of the fixed-family NLME study.

Conditional prediction

Two hundred independently generated $N = 300$ datasets used Gamma CRP, Normal albumin and CRP–clearance score correlation 0.55. A stock Gaussian NLME fit supplied only finite initial values; Gaussian and flexible-margin FREM were then fitted afresh for 1,500 score iterations. Conditional-median clearance was evaluated on a fixed CRP grid, with pointwise 2.5th–97.5th percentiles across fitted datasets. For each fit, a common-random-number population of 30,000 subjects above the 95th CRP percentile supplied the 10th, 50th and 90th response percentiles at seven times; albumin was integrated out. Predictive error was log-RMSE against generating quantiles.

Numerical verification

Analytical complete-data scores, Fisher's identity, categorical augmentation, moving-support paths, the Gaussian nested model and the FREM-to-FFEM mapping were checked against independent calculations. Because Gamma shape and scale can compensate while retaining the marginal mean, a separate direct-MLE experiment used 10,000 independently generated Gamma samples at each of $N = 50$, 100, 300, 1,000 and 3,000. Full-model Gamma estimates were also compared with direct Gamma MLEs on the same $N = 100$ and $N = 300$ datasets.

RESULTS

Fixed-family re-estimation

Table 1: Flexible-margin FREM parameter recovery and difference in estimated joint observed-data log likelihood relative to Gaussian FREM. Bias is relative mean bias across 100 datasets; $\Delta\ell$ is flexible-margin minus Gaussian FREM.

Scenario	V bias (%)	CL bias (%)	Residual bias (%)	$\Delta\ell$
Normal/Normal	0.29	-0.03	0.03	-0.06
lognormal/Normal	-0.03	-0.46	-0.15	4.30
Gamma/Normal	0.31	0.18	-0.41	4.60
Weibull/lognormal	-0.26	0.01	0.05	9.41

Structural RMSE was nearly identical between methods. Marginal parameters were close to generating values: estimated lognormal sdlog values were 0.239 and 0.277 (truth 0.240 and 0.280), and Weibull shape/scale were 2.309/78.11 (truth 2.3/78). Dataset-specific Gamma estimates matched direct MLEs within mean absolute differences 0.0065 (shape) and 0.0101 (scale). Maximum absolute mean correlation bias was 0.022.

Flexible margins had larger joint likelihood in 285/300 non-Gaussian datasets. Most mean improvement arose from the covariate-density component (4.01, 4.15 and 8.58), with conditional-response contributions 0.29, 0.45 and 0.83. The models are non-nested and the independent importance-sampling comparisons are descriptive; by (8a), joint improvement alone does not establish improved $p(Y \mid c)$.

Automatic eta-margin selection

Table 2: Agreement of posterior eta-margin selection with the analytically available direct observed-likelihood ranking. Agreement concerns the ranking, not necessarily recovery of the generating family in a finite dataset.

Information	Generating family	Datasets	Ranking agreement
Informative	Normal	12	12/12
Informative	Laplace	12	12/12
Weak, 2,000 draws	Normal or Laplace	24	22/24
Weak, targeted 20,000-draw rerun	Discordant subset	2	2/2

In informative data, posterior log-likelihood RMSE was below 0.001 for Normal and 0.071 for Laplace candidates. Under weak information, the two initial discordances had small direct advantages (0.164 and 0.114) and poor overlap; 20,000 draws recovered both direct rankings. Automatic decisions were therefore accepted only when ranking and overlap diagnostics were resolved.

Conditional prediction

Mean V, CL and residual estimates were nearly identical between arms (10.021, 3.014 and 0.120). At the 99.9th CRP percentile, generating median clearance was 5.44. Across fits, the flexible median was 5.28 with a pointwise 95% interval of 4.75–5.90, whereas Gaussian FREM gave 7.60 (6.41–9.36; Figure 1A).

Flexible margins had lower conditional VPC log-RMSE in 174/200 paired fits. Median RMSE was 0.085 versus 0.336 for Gaussian FREM; the mean paired difference was 0.246 (95% CI 0.219–0.272), and the median relative reduction was 74.4% (Figure 1B).

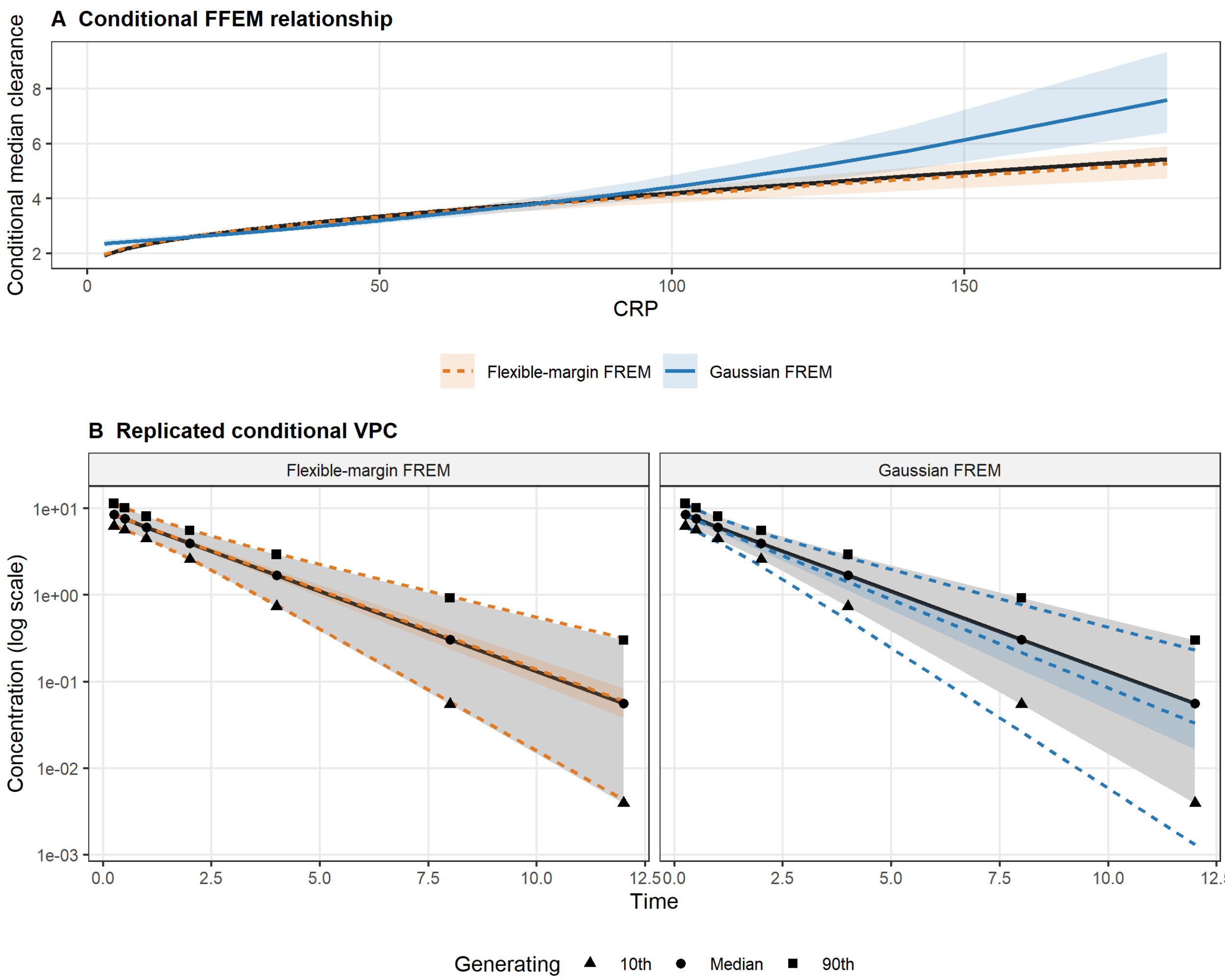

Figure 1: Conditional results across 200 independently fitted datasets. (A) Median FFEM/conditional relationships; ribbons are pointwise 2.5th–97.5th percentiles across fits and the black line is the generating relationship. (B) Replicated conditional VPC. Grey regions and black symbols show the generating 10th (triangle), 50th (circle) and 90th (square) percentiles. Coloured dashed lines are median fitted quantiles across the 200 fits; coloured ribbons are pointwise 2.5th–97.5th percentiles for the fitted median.

Numerical verification

Analytical scores agreed with independent derivatives within 6.5×10^{-8}; Fisher discrepancies were below their Monte Carlo error, including multiple categorical and moving-support examples. Gamma shape bias decreased from 2.83% at $N = 100$ to 0.14% at $N = 3,000$, while full-model and same-dataset direct MLEs agreed within 0.0075 at $N = 300$. Gaussian FREM block formulas, non-Normal conditional draws and direct-versus-translated FFEM predictions agreed to the stated numerical tolerances in the reproducibility tests.

DISCUSSION

The method removes the need to know latent eta distributions in advance. A Normal-eta incumbent supplies full posterior draws, candidate families are ranked outside the recursion, and the selected fixed model is fitted afresh. This preserves one likelihood within each fit and avoids treating shrunken conditional modes as population observations. Rankings matched the direct likelihood whenever Monte Carlo and overlap gates resolved. In the separate fixed-family NLME study, structural recovery remained accurate and non-Gaussian margins increased joint likelihood without changing the response model.

The clearest practical difference was prediction conditional on covariate-tail values. A misspecified marginal CDF changes the Gaussian score assigned to a covariate value and therefore changes the fitted distribution $p(\eta_{\text{par}} \mid c)$ even when the first two moments are estimated accurately. Across 200 fits, flexible margins substantially reduced conditional VPC-quantile error and contained the generating upper-tail relationship within their between-fit band.

The conditional representation is a conventional transformed-covariate FFEM for Normal parameter margins and an exact quantile-transformed generative model otherwise. Direct and translated predictions coincide at fixed population parameters; re-estimation of the translation instead targets $p(y \mid c_S)$.

The controlled-Markov score recursion changes the computation, not the target: Fisher's identity gives the observed-likelihood gradient. Convergence remains conditional on Appendix B's assumptions. Automatic screening is likewise diagnosed rather than unconditional: Jensen bias favours the incumbent, and heavy-tailed candidates may require more draws, defensive proposals or direct refitted-likelihood comparison despite an apparently adequate finite ESS.

The evaluation used one PK model; automatic selection was benchmarked in a simpler latent model. Larger nonlinear selection and coverage studies remain necessary. General non-Gaussian R-vines, nonregular support-boundary MLEs, permutation-invariant nominal models and fixed-only estimated structural parameters are outside scope. Supported margins must be proper, identifiable and satisfy the stated differentiability and moment conditions; this is not a distribution-free estimator.

CONCLUSION

Flexible-margin Gaussian-copula FREM separates marginal distributions from dependence and reduces to Gaussian FREM when all margins are Normal. Latent eta families need not be known in advance: a Normal-eta incumbent, full-posterior screen, diagnosed independent validation and fresh selected-family fit form one coherent likelihood workflow. For every fixed family specification, Fisher's identity links the controlled-Markov stochastic-approximation mean field to the observed-data likelihood score. In the simulations considered here, structural-parameter recovery remained accurate, while flexible margins improved the joint likelihood and conditional tail prediction under non-Gaussian generating distributions. The fitted conditional model translates exactly to a conventional FFEM when parameter margins are Normal and to an exact conditional generative representation otherwise.

DECLARATIONS

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APPENDIX A: Likelihood normalization, Fisher identity and FFEM translation

Continuous normalization

For fixed θ, the map $x_j \mapsto \xi_j = \Phi^{-1}\{F_j(x_j)\}$ has Jacobian $d\xi_j/dx_j = f_j(x_j)/\phi(\xi_j)$. Substitution in (7) therefore gives

$$p_{\theta_p}(x)\,dx = \phi_R(\xi)\,d\xi.$$

Integration over $\mathbb{R}^d$ is one. This proves normalization for any proper absolutely continuous margins; Gaussian marginal shape is not required.

Several categorical covariates

For observed category vector $k_{\mathcal{D}}$, let $u_j = b_{j,k_j-1} + v_j\pi_{j,k_j}$ for $j \in \mathcal{D}$. The augmented mixed density is

$$p_\theta^{\text{aug}}(x_{\mathcal{C}}, k_{\mathcal{D}}, v_{\mathcal{D}}) = c_R(u)\prod_{j\in\mathcal{C}} f_j(x_j)\prod_{j\in\mathcal{D}} \pi_{j,k_j}. \tag{A1}$$

Since $du_j = \pi_{j,k_j}dv_j$, integrating $v_{\mathcal{D}} \in (0,1)^{|\mathcal{D}|}$ gives the Gaussian-copula rectangle mass for $k_{\mathcal{D}}$. Summing category vectors tiles the latent unit cube, so the masses sum to one. Missing categories add counting-measure summation and remain normalized.

Moving continuous support

For $X = F_\alpha^{-1}(W)$ with $W \in (0,1)$, $dW = f_\alpha(X)dX$. Thus a latent moving-support coordinate contributes the fixed reference dW and its marginal density factor disappears after change of variables. Dependence still enters through $\xi = \Phi^{-1}(W)$. If the coordinate affects the response, $p_\theta\{Y \mid h^{-1}(C\mu + F_\alpha^{-1}(W))\}$ is differentiated through both μ and F_α^{-1}. The moving-boundary derivative is therefore represented by the pathwise quantile derivative rather than discarded.

Observed likelihood and Fisher identity

Multiplying the normalized population density by the conditional response density and integrating over all unobserved coordinates gives $g_i(\theta)$. Under an integrable derivative envelope,

$$\begin{aligned} E_\vartheta[\nabla \log \widetilde{f}_i(Z_i;\vartheta) \mid O_i] &= \int \frac{\nabla \widetilde{f}_i(z;\vartheta)}{\widetilde{f}_i(z;\vartheta)}\,\frac{\widetilde{f}_i(z;\vartheta)}{\widetilde{g}_i(\vartheta)}\,\lambda(dz) \\ &= \frac{\nabla \int \widetilde{f}_i(z;\vartheta)\lambda(dz)}{\widetilde{g}_i(\vartheta)} = \nabla \log \widetilde{g}_i(\vartheta). \end{aligned}$$

The chain rule gives $\nabla_\vartheta \bar{\ell} = J_\tau(\vartheta)^\mathsf{T}\nabla_\theta \bar{\ell}$. A nonsingular J_τ makes native- and internal- coordinate stationary points equivalent.

Gaussian nested model and FFEM translation

When every margin is Normal, $X = \mu + D\xi$ with $\xi \sim N_d(0,R)$, hence $X \sim N_d(\mu, DRD)$ and $\Omega_{\text{FREM}} = DRD$ exactly.

Let P index the parameter random effects and let S be any selected subset of continuous covariates. Marginalizing covariates outside S leaves the Gaussian subvector (ξ_P, ξ_S) with the corresponding submatrix of R. The multivariate-Normal conditioning formula gives

$$\xi_P \mid \xi_S = z_S = B_{\xi,S}z_S + L_S\varepsilon, \qquad B_{\xi,S} = R_{P,S}R_{S,S}^{-1}, \qquad L_S L_S^\mathsf{T} = R_{P,P} - B_{\xi,S}R_{S,P}, \tag{A2}$$

where $\varepsilon \sim N(0,I)$. If the parameter margins are Normal, $\eta_P = D_P\xi_P$; substitution in (A2) yields the coefficient matrix $D_P B_{\xi,S}$ and UPV covariance $D_P L_S L_S^\mathsf{T} D_P$ stated in the conventional FFEM representation. If the selected covariate margins are also Normal, replacing z_S by $D_S^{-1}(c_S - \mu_S)$ gives the usual raw-scale FREM coefficient matrix and Schur-complement UPV covariance.

For continuous non-Normal parameter margins, let Q_η apply their quantile functions componentwise. Since $\eta_P = Q_\eta\{\Phi(\xi_P)\}$, (A2) gives

$$\eta_P \mid c_S = Q_\eta\left[\Phi\left\{B_{\xi,S} z_S(c_S) + L_S \varepsilon\right\}\right]. \tag{A3}$$

Equation (A3) is the exact non-Normal conditional representation in the main text and has the same conditional law as the fitted FREM. For an ordered categorical covariate, conditioning is instead on its latent Gaussian interval; integrating (A2) over the corresponding truncated-Normal score distribution gives the stated interval mixture. Finally, direct FREM and translated-FFEM predictions integrate the same response kernel $p(Y \mid \eta_P)$ against this same conditional distribution. Their conditional predictive distributions are therefore identical when the population parameters are held fixed.

Posterior likelihood-ratio identity

Let f_{0i} and f_{mi} be incumbent and candidate complete-data densities with respect to the same fixed measure λ, and assume $f_{mi} = 0$ wherever the ratio is otherwise undefined. Since $\pi_{0i} = f_{0i}/g_{0i}$,

$$\begin{aligned} E_{\pi_{0i}}\left(\frac{f_{mi}}{f_{0i}}\right) &= \int \frac{f_{mi}(z)}{f_{0i}(z)} \frac{f_{0i}(z)}{g_{0i}} \lambda(dz) \\ &= \frac{1}{g_{0i}} \int f_{mi}(z)\, \lambda(dz) = \frac{g_{mi}}{g_{0i}}. \end{aligned} \tag{A4}$$

When only the eta distribution changes, the response density cancels, giving (14). The Normal incumbent dominates the screened families, but (A4) alone does not ensure finite variance or adequate mixing; Student ratios may have infinite second moments. Independent fitting/validation pools separate optimization noise, after which the pools are discarded and the selected model is fitted afresh.

APPENDIX B: Controlled kernel, convergence and numerical error

Exact conditional-population proposal

Let $q_{\vartheta,i}(\psi)$ denote the population density of ψ_i conditional on observed covariates, from which proposals are sampled exactly, and let $L_{\vartheta,i}(\psi) = p_\theta(Y_i \mid \psi, x_i)$. Then

$$\pi_{\vartheta,i}(\psi) = \frac{L_{\vartheta,i}(\psi) q_{\vartheta,i}(\psi)}{\int L_{\vartheta,i}(v) q_{\vartheta,i}(v) dv}.$$

Suppose uniformly on the compact parameter set that $L_{\vartheta,i}(\psi) \le M_i$ and its normalizer is at least $m_i > 0$. The independence-Metropolis weight $w = \pi/q$ is bounded by M_i/m_i, and therefore

$$P^{\text{ind}}_{\vartheta,i}(\psi, A) \ge (m_i/M_i)\Pi_{\vartheta,i}(A). \tag{B1}$$

Composition with any other invariant Metropolis kernels preserves the minorization. With finitely many subjects and chains, the product chain has a uniform Doeblin bound. Exact conditional augmentation after the eta move also preserves it: if $R_\vartheta(da \mid \psi')$ draws missing/categorical/fixed- reference variables, then

$$\widetilde{P}_\vartheta\{(\psi, a), d\psi' da'\} = P_\vartheta(\psi, d\psi') R_\vartheta(da' \mid \psi') \ge \epsilon \Pi_\vartheta(d\psi', da'). \tag{B2}$$

If a rejection sampler for a multivariate category rectangle reaches its finite iteration limit, the iteration is terminated without substituting an approximate draw.

Markovian stochastic-approximation decomposition

After finite adaptation, write the score noise as the conditional mean plus a martingale term, a Poisson-equation remainder and numerical error:

$$\widehat{H}_{k+1} = \nabla \bar{\ell}(\vartheta_k) + e_{k+1} + r^{\text{P}}_{k+1} + r^{\text{num}}_{k+1}. \tag{B3}$$

Here is the precise mapping to controlled-Markov stochastic approximation [19, 20, 14, 15]. On every visited compact parameter set, assume their H1–H2 with a drift function W: the partially marginalized score is uniformly W-bounded, each P_ϑ has unique invariant distribution Π_ϑ, the kernels are uniformly W-geometrically ergodic, and, for some $p \ge 2$,

$$\sup_\vartheta P_\vartheta W^p \le \rho W^p + b, \qquad \rho < 1. \tag{B4}$$

The common minorization in (B1)–(B2), together with Lipschitz dependence of the kernel after finite adaptation, supplies H3 with exponent $\upsilon = 1$. H4 is the following integrated local-oscillation condition for the actual fixed-preconditioner increment $\mathcal{H}_\vartheta$: for every compact K and $\delta > 0$,

$$\sup_{\vartheta \in K} \int \Pi_\vartheta(dx) \sup_{\substack{\vartheta' \in K \\ \|\vartheta' - \vartheta\| \leq \delta}} \|\mathcal{H}_{\vartheta'}(x) - \mathcal{H}_\vartheta(x)\| \leq C_K \delta^\alpha, \qquad \alpha = 1. \tag{B5}$$

Smoothness of the mean field alone does not imply (B5); it is a stated regularity condition. Under H1–H4, the associated Poisson-equation solutions have the bounds used by Fort et al. Proposition 4.11. Their H6 requires a power-law gain exponent satisfying

$$\max\left\{\frac{1}{p}, \frac{1 + (\alpha \wedge \upsilon)/p}{1 + (\alpha \wedge \upsilon)}\right\} < a \leq 1. \tag{B6}$$

For $p \geq 2$ and $\alpha = \upsilon = 1$, the implemented $a = 0.80 > 0.75$ satisfies (B6), not merely the usual Robbins–Monro sums.

After safeguards become inactive, apply Fort et al. Proposition 4.6 to the stopped tail recursion with mean field $h = A\nabla\bar{\ell}$. Its C-i follows from differentiability, C-ii from compact-tail stability, and C-iii from the gain. For H5 and C-v(A), assume a coercive continuously differentiable extension w that equals $-\bar{\ell} + C$ near the visited compact tail, has compact level sets and compact zero-drift set

$$\mathcal{L}_w = \{\vartheta : \langle \nabla w(\vartheta), h(\vartheta) \rangle = 0\},$$

such that $w(\mathcal{L}_w)$ has empty interior and $\mathcal{L}_w \cap K_{\text{tail}} = \mathcal{L} \cap K_{\text{tail}}$. This global extension and its critical-value property are assumptions, not automatic consequences of local likelihood smoothness.

For Proposition 4.6 C-iv, apply Proposition 4.11 and Corollary 4.12 to the process stopped on exit from the compact tail. For $A_k = \sup_{l \geq k} \| \sum_{j=k}^{l} \gamma_j \xi_j \|$, their result gives $P(A_k \geq \delta/2) \to 0$; because $A_n \leq 2A_k$ for $n \geq k$, continuity from above and countable rational $\delta > 0$ give $A_k \to 0$ almost surely on the eventual no-exit tail. Proposition 4.6 therefore yields $d(\vartheta_k, \mathcal{L}) \to 0$ almost surely. The finite prefix cannot change the limit set. This is a conditional stationary-set result for the Younes stochastic-gradient recursion presented as Delyon et al. equation (74), not an invocation of their finite-statistic SAEM Theorem 5 or 8.

For centered numerical scores, suppose $\|r_k(\vartheta, u)\| \leq h_k^2 M(u)$ with $\sup_k E\{M(Z_k)\} < \infty$. Taking $h_k = h_0\sqrt{\gamma_k}$ gives $E\sum_k \gamma_k \|r_k\| \leq C h_0^2 \sum_k \gamma_k^2 < \infty$; hence the complete numerical perturbation is almost surely summable. Post-freeze one-sided differences are rejected because they lack this second-order bound.

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